

UniTS Anomaly Detection Report

ESA-Mission1 | Generated 2026-04-24 00:26

Dataset Summary

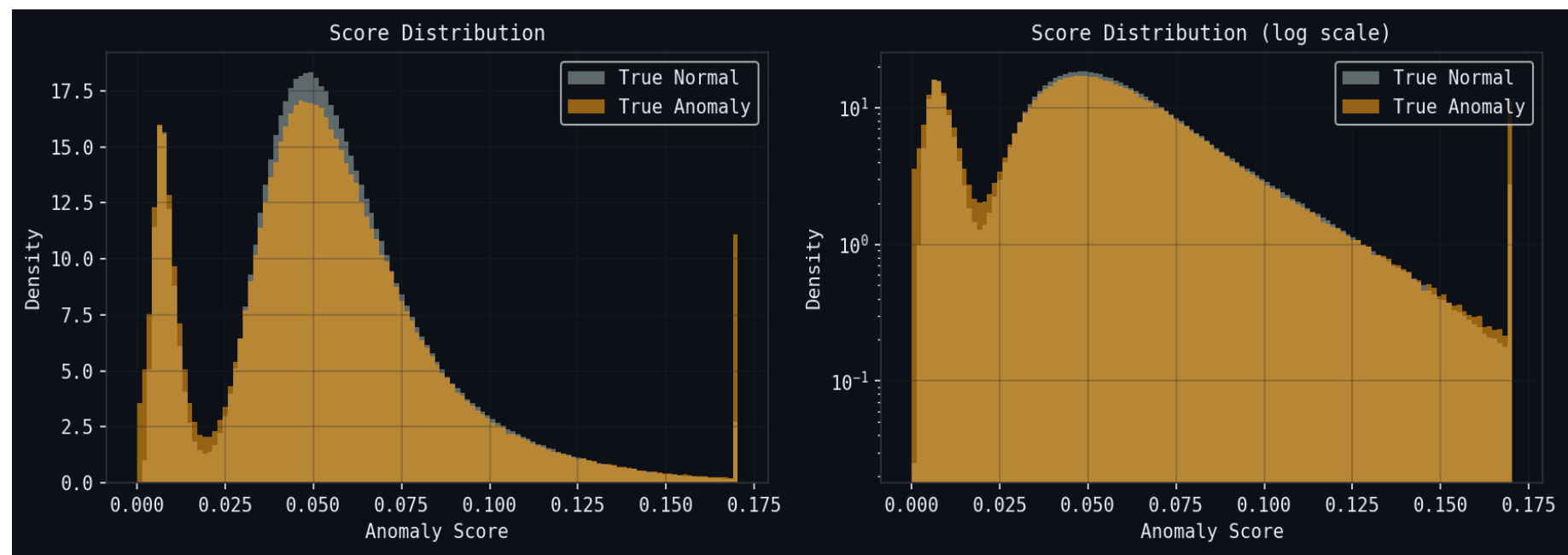
Total timesteps	7,364,160
Date range	2007-01-01 to 2013-12-31
Ground truth anomaly rate	11.10% (817,289 points)
Predicted anomaly rate	10.83% (797,344 points)

Detection Metrics

Metric	Value	Interpretation
Precision	0.9937	Of all predicted anomalies, 99.4% are real
Recall	0.9695	96.9% of all true anomaly points are detected
F1 Score	0.9814	Harmonic mean of precision and recall
Accuracy	0.9959	Overall correct classification rate
ROC-AUC	0.4923	Raw score discrimination (near-random — see note below)
Avg Precision	0.1246	Area under the Precision-Recall curve
True Positives	792,325	Anomaly points correctly flagged
False Positives	5,019	Normal points incorrectly flagged (0.1% of all normal points)
False Negatives	24,964	Anomaly points missed
True Negatives	6,541,852	Normal points correctly cleared

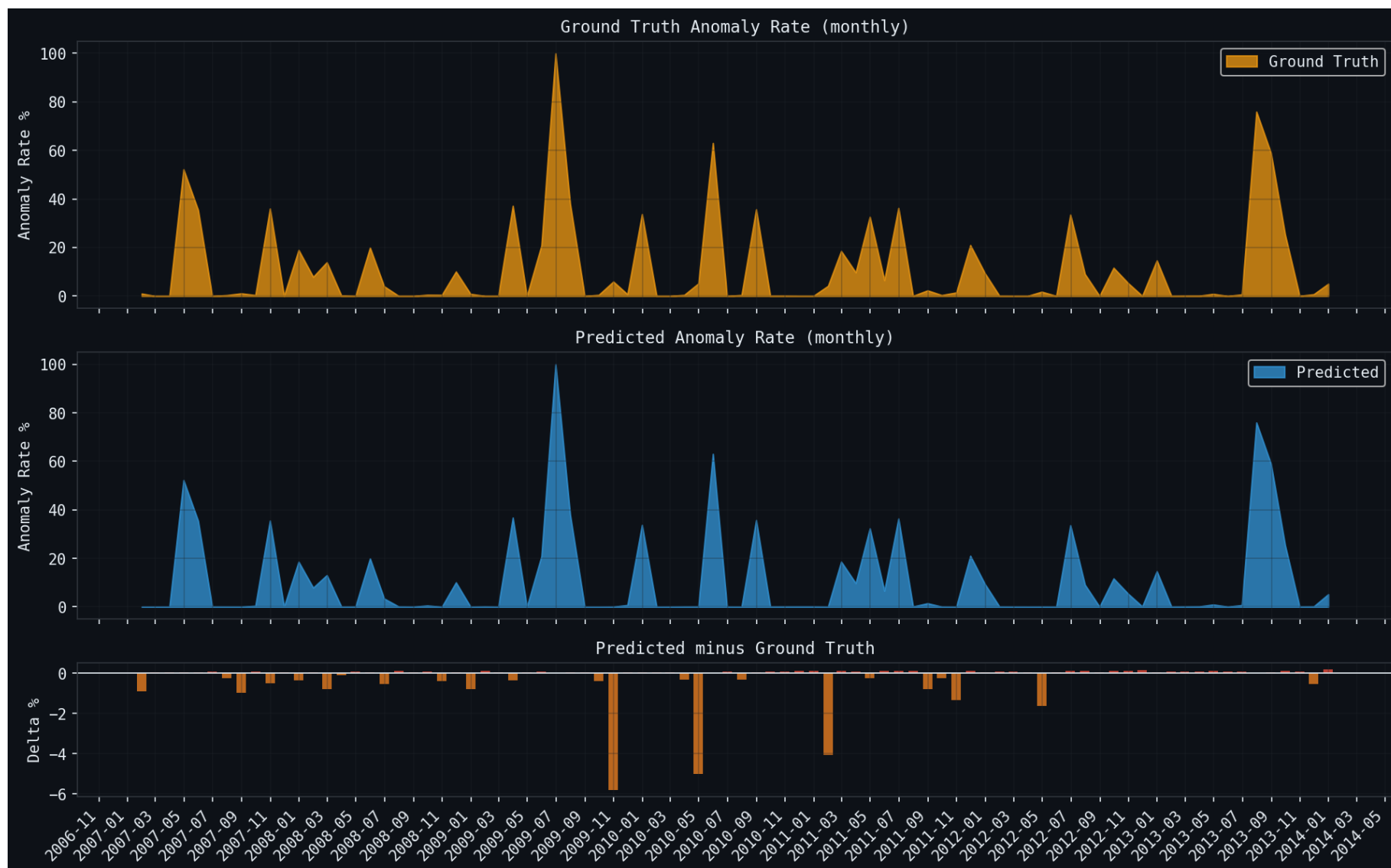
Note on F1 vs ROC-AUC: The F1 of 0.981 reflects binary detection quality under the UniTS point-adjust (PA) evaluation procedure, where detecting any point within an anomaly segment credits the full segment. The ROC-AUC of 0.492 reflects the raw anomaly_score used as a continuous ranking signal — it is near-random, meaning scores should not be used to prioritize or rank alerts. The binary flag is the actionable output.

Score Distribution



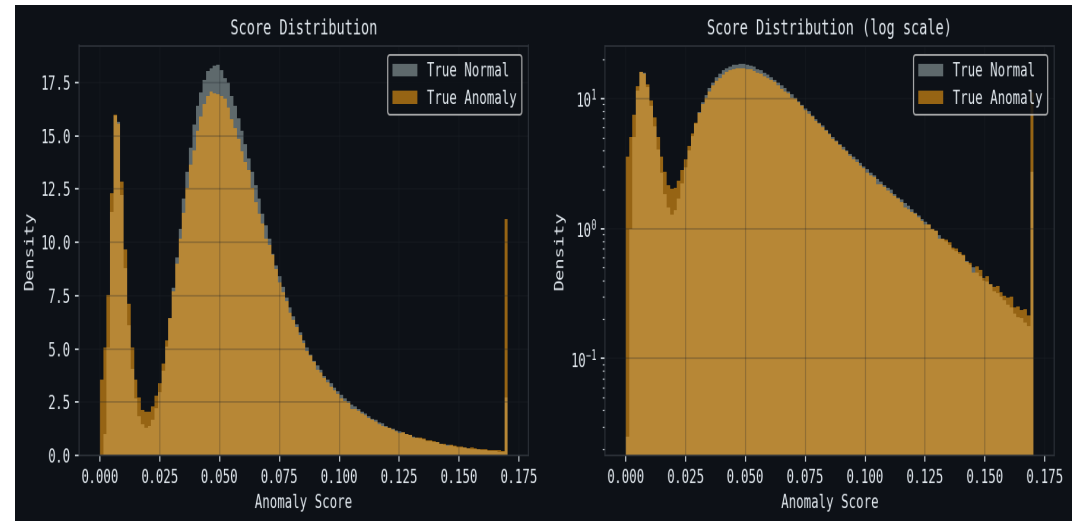
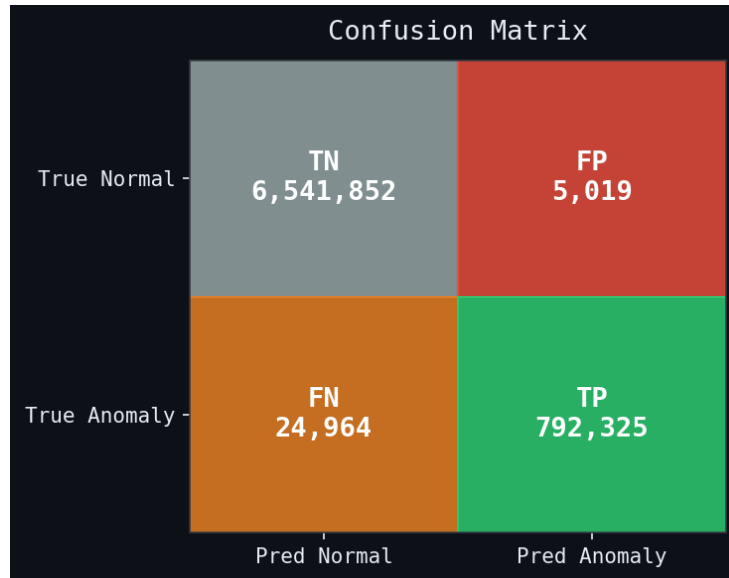
Score distributions clipped at the 99.5th percentile. Overlap between True Normal and True Anomaly distributions indicates the model relies on a fixed threshold rather than separating score distributions — consistent with ROC-AUC of 0.492.

Anomaly Timeline (monthly)

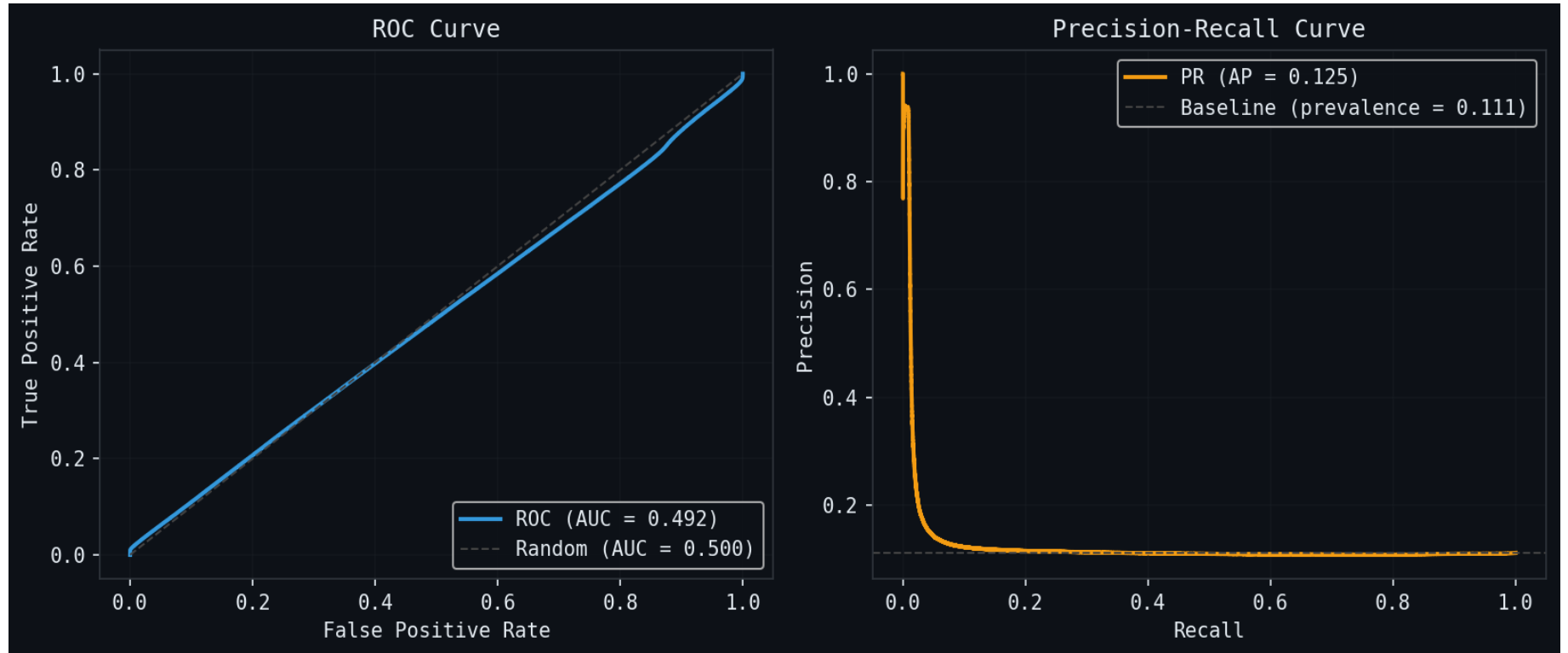


Most months show predicted rate slightly above ground truth, consistent with the 5,019 false positives in the test period. Predicted and ground truth anomaly rates track closely through all major anomaly events.

Confusion Matrix & Score Distribution



ROC & Precision-Recall Curves



Both curves are computed on the raw anomaly_score (continuous). ROC-AUC = 0.492 (near-random). The binary is_anomaly_predicted flag (F1 = 0.981) is the operationally useful output.

Top False Positive Windows (pred=1, truth=0)

Total FP windows: 4,274 — showing top 20 by duration.

Start	End	Duration (pts)	Peak Score
2009-05-25 12:48:00	2009-05-25 13:09:00	43	686251.1000
2011-02-23 19:26:30	2011-02-23 19:41:00	30	385.8235
2007-04-14 08:00:00	2007-04-14 08:12:30	26	36.6819
2011-11-24 17:06:30	2011-11-24 17:17:30	23	439.6085
2012-06-20 22:35:00	2012-06-20 22:45:00	21	163.7499
2011-02-23 19:12:00	2011-02-23 19:21:00	19	76.0220
2012-06-20 22:24:30	2012-06-20 22:32:30	17	4.5992
2013-09-12 01:52:00	2013-09-12 01:59:30	16	11.8357
2011-11-24 16:48:00	2011-11-24 16:55:30	16	9.2418
2011-06-09 02:50:00	2011-06-09 02:57:00	15	39.5455
2011-06-09 02:31:00	2011-06-09 02:38:00	15	2.3862
2011-11-24 16:56:30	2011-11-24 17:03:00	14	11.9380
2011-06-09 02:41:00	2011-06-09 02:46:00	11	2.7851
2011-06-09 02:25:00	2011-06-09 02:30:00	11	1.2168
2010-08-18 22:27:00	2010-08-18 22:31:30	10	9.4190
2013-09-12 01:39:30	2013-09-12 01:43:30	9	6.9940
2011-02-23 19:22:00	2011-02-23 19:25:30	8	17.2433
2009-01-15 23:08:30	2009-01-15 23:11:30	7	0.7367
2009-03-13 03:07:30	2009-03-13 03:10:30	7	197.8406
2009-01-15 22:57:30	2009-01-15 23:00:30	7	3.7404

Top False Negative Windows (pred=0, truth=1)

Total FN windows: 34 — showing top 20 by duration.

Start	End	Duration (pts)	Peak Score
2009-10-13 06:39:30	2009-10-15 02:27:00	5,256	0.1944
2010-04-28 06:00:30	2010-04-29 18:19:30	4,359	0.2106
2011-01-27 03:53:00	2011-01-27 19:12:00	1,839	0.1958
2011-01-28 22:29:30	2011-01-29 13:47:00	1,836	0.2083
2012-04-20 22:09:30	2012-04-21 10:29:00	1,480	0.2019
2011-10-08 07:09:00	2011-10-08 17:33:30	1,250	0.2062
2007-08-14 22:15:00	2007-08-15 05:56:30	924	0.1736
2007-01-12 15:42:00	2007-01-12 22:47:30	852	0.1653
2008-12-26 21:13:30	2008-12-27 03:19:00	732	0.1815
2008-02-21 11:37:30	2008-02-21 17:28:00	702	0.2032
2011-08-16 14:53:00	2011-08-16 20:25:30	666	0.2043
2013-11-23 06:52:30	2013-11-23 11:20:30	537	0.1973
2007-10-26 12:21:00	2007-10-26 16:32:00	503	0.1048
2008-06-10 16:20:30	2008-06-10 19:44:00	408	0.0634
2009-03-08 10:33:00	2009-03-08 13:53:00	401	0.0389
2008-10-15 20:24:00	2008-10-15 23:38:00	389	0.0528
2010-03-07 11:29:30	2010-03-07 14:33:30	369	0.1711
2010-07-31 08:42:30	2010-07-31 11:30:00	336	0.1861
2007-12-24 02:30:30	2007-12-24 05:17:30	335	0.0925
2009-09-21 09:17:00	2009-09-21 12:03:00	333	0.1389